WHEN TO ADD VASOPRESSIN

Septic shock | VASSPR trial | Pressor selection and physiology

RECOGNIZE → THINK PHYSIOLOGY → ACT → REASSESS

THE BEDSIDE PROBLEM

Septic vasoplegia reduces systemic vascular resistance. Norepinephrine restores vascular tone, but escalating catecholamines can increase arrhythmia burden and myocardial oxygen demand.

Vasopressin stimulates vascular V1 receptors independently of adrenergic signalling. It is an adjunct to norepinephrine, not a substitute for diagnosing why shock persists.

BEFORE ESCALATING VASOPRESSORS

• Confirm source control, antimicrobials, adequate oxygen delivery and a reliable arterial pressure measurement.

• Assess fluid responsiveness rather than reflexively giving more crystalloid; use clinical perfusion, dynamic assessment and serial POCUS.

• Look for mixed shock: LV dysfunction, RV failure/PE, tamponade, tension pneumothorax, haemorrhage and dynamic LVOTO.

• Reassess skin perfusion, mentation, lactate trend and urine output; MAP alone is not the endpoint.

PRACTICAL ED SEQUENCE

1 Norepinephrine first-line; titrate to a usual initial MAP target around 65 mmHg, individualized to the patient.

2 If vasoplegia persists or the norepinephrine requirement rises, consider adding fixed-dose vasopressin (commonly 0.03 units/min) according to local policy.

3 If shock persists, reassess haemodynamics and the diagnosis; do not use a catecholamine threshold alone as the decision rule.

Distinguish a clinical starting framework from a validated universally optimal vasopressin threshold.

WHAT VASSPR ACTUALLY RANDOMIZED

Design: Pragmatic cluster-randomized, monthly crossover trial at 13 US hospitals; 3,959 adults with septic shock.

Strategies: Recommend vasopressin when other vasopressors reached 0.1 versus 0.4 µg/kg/min norepinephrine-equivalent.

Important: This was a policy comparison, not random assignment of each individual patient to one exact drug dose.

PRIMARY OUTCOME: 28-DAY DEATH

Earlier-threshold strategy: 628/1,975 (31.8%).

Later-threshold strategy: 655/1,984 (33.0%).

Adjusted absolute risk difference −0.4 percentage points (95% CI −2.7 to +1.3); P=0.62.

There was no demonstrated mortality advantage to the earlier hospital-level recommendation.

WHAT THIS RESULT DOES NOT MEAN

• It does not prove the two initiation thresholds are clinically equivalent in every phenotype of septic shock.

• It does not mandate waiting for 0.4 µg/kg/min before adding vasopressin.

• Actual treatment overlapped: median norepinephrine-equivalent dose before vasopressin was 0.14 versus 0.34 µg/kg/min.

• The cluster design and incomplete separation of actual treatments limit an individual-patient threshold inference.

THE ONE-MINUTE TAKE-HOME

Vasopressin is reasonable for persistent vasodilatory shock, but VASSPR did not identify a mortality-improving early trigger.

Treat the phenotype and trajectory. If MAP is adequate but perfusion remains poor, ask whether the problem is cardiac output, venous congestion or an unaddressed source.

Peltan ID et al. JAMA. 2026. DOI 10.1001/jama.2026.21333 | Surviving Sepsis Campaign, Intensive Care Med. 2021. DOI 10.1007/s00134-021-06506-y

Evidence: VASSPR full-text results reviewed; guideline recommendations are distinct from trial findings. Canadian practice: follow local protocols.

WHOLE BLOOD BEFORE HOSPITAL

Traumatic haemorrhagic shock | Low-titre group O blood | Transport priorities

RECOGNIZE → THINK PHYSIOLOGY → ACT → REASSESS

FIRST PRINCIPLES: STOP THE BLEEDING

Exsanguination requires haemorrhage control and haemostatic resuscitation in parallel. A rise in blood pressure after transfusion is not definitive haemorrhage control.

• Compress/pack external bleeding; apply tourniquet or pelvic binder when indicated.

• Identify chest/abdominal/pelvic sources; expedite surgery or interventional radiology.

• Prevent hypothermia; reassess ionized calcium and acid-base physiology during massive transfusion.

WHY LOW-TITRE O WHOLE BLOOD?

A single unit delivers red cells, plasma and platelets together, simplifying early balanced transfusion and reducing multiple-component logistics.

Low anti-A/anti-B titre is used to reduce haemolysis risk during emergency non-group-specific transfusion; blood bank policy determines eligibility and handling.

O-positive versus O-negative selection has recipient-specific Rh considerations, especially for patients with childbearing potential.

PREHOSPITAL RECOGNITION AND ACTIONS

• Suspect life-threatening haemorrhage with penetrating injury, uncontrolled external bleeding, concerning mechanism, poor perfusion or escalating shock.

• Shock index = heart rate / systolic BP (in mmHg). An elevated value supports concern, but it can be confounded by drugs, age and physiology.

• Start available blood according to service protocol while maintaining rapid transport to definitive haemorrhage control.

• Recheck vitals and mental status repeatedly: transient improvement can mask ongoing bleeding.

WHAT THE 2026 IMPLEMENTATION STUDY FOUND

Design: Retrospective descriptive program evaluation of EMS low-titre O-positive whole blood in Oklahoma City and Tulsa (Dec 2022–Feb 2025).

Case mix: Penetrating trauma accounted for 71.6% of transfused cases.

Physiologic response: Systolic BP, heart rate and shock index improved after transfusion (reported $P<0.05$).

A rise in systolic BP was associated with survival to hospital arrival: OR 9.7 (95% CI 1.6–58.6).

CRITICAL APPRAISAL — AVOID CAUSAL CLAIMS

• No randomized control group: no proof that whole blood was better than balanced component therapy or that it improved mortality.

• Physiologic improvement is vulnerable to confounding and survivor bias; the wide odds-ratio CI indicates imprecision.

• This evaluated an urban EMS system with defined logistics; external applicability depends on storage, transport and transfusion protocols.

• Patients can deteriorate again en route — blood may buy time, not stop bleeding.

ED HANDOVER AND NEXT MOVES

State units already given, product type, timing, estimated blood loss, haemodynamics and response.

Activate the local massive-haemorrhage pathway when indicated. Coordinate compatibility testing, additional balanced products, fibrinogen and calcium assessment, warming, and immediate source control.

Do not delay definitive treatment waiting for normal blood pressure, lactate or a “complete” laboratory panel.

THE ONE-MINUTE TAKE-HOME

Early whole blood is a feasible haemostatic bridge; better vitals after transfusion ≠ definitive control or proven survival benefit.

The decisive question is whether the patient is moving rapidly toward a haemostasis-capable destination and intervention.

Bailey AM et al. Prehosp Emerg Care. 2026. DOI 10.1080/10903127.2026.2747040 | Original abstract reviewed (PMID 42896084).

Observational evidence only. Apply Alberta massive-haemorrhage, transport and blood-bank protocols.